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Re: some bioinformatic problems from Changchun Xiao lab

发件人：Kylie Shen <kylie@eclipsebio.com>
时 间：2019年8月14日（星期三）上午0：19
收件人：陈鹏达 <pandaman5555@qq.com>
抄 送：Eric Van Nostrand <eric@eclipsebio.com>; 肖昌春 <cxiao@xmu.edu.cn>; alex <alex@eclipsebio.com>; 廖堃玉 <779223993@qq.com>

纯文本 |

邮件可翻译为中文 立即翻译

Hi Pengda,

That should be fine. I’m posting the links below as well, in case you want to take a look at those. Also, note that these rRNA sequences are included in the file that I sent you previously.

28S rRNA: https://www.ncbi.nlm.nih.gov/gene/236598

18S rRNA: https://www.ncbi.nlm.nih.gov/gene/19791

5S rRNA: https://www.ncbi.nlm.nih.gov/gene?Db=gene&Cmd=DetailsSearch&Term=100169751

5.8S rRNA: https://www.ncbi.nlm.nih.gov/gene/790956

45S rRNA: https://www.ncbi.nlm.nih.gov/nucleotide/NR_046233.2?report=genbank&log$=nucltop&blast_rank=4&RID=G62RJ921015

Thank you,
Kylie

Kylie Shen, Bioinformatician
Eclipse BioInnovations
kylie@eclipsebio.com

On Aug 12, 2019, at 6:44 PM, 陈鹏达 <pandaman5555@qq.com> wrote:

Hi Kylie,

I'm very grateful for your help.
For rRNA, we used to have a mm10 rRNA sequence from UCSC. Is it okay to use this for our aim? Or please give a link of the rRNA sequences on NCBI you mentioned in the email.

Thank you!
Pengda

------------------ 原始邮件 ------------------

发件人: "Kylie Shen"<kylie@eclipsebio.com>;
发送时间: 2019年8月13日(星期二) 凌晨5:46
收件人: "陈鹏达"<pandaman5555@qq.com>;
抄送: "Eric Van Nostrand"<eric@eclipsebio.com>;"肖昌春"<cxiao@xmu.edu.cn>;"alex"<alex@eclipsebio.com>;"廖堃玉"<779223993@qq.com>;
主题: Re: some bioinformatic problems from Changchun Xiao lab

Hi Pengda,

We also don't have a subscription to Repbase, so I have been using repetitive elements pulled from DFAM: https://dfam.org/home. We also found the rRNA sequences on NCBI, and I can send you the links to those pages if you would like. I'm attaching the mouse specific version that I use, in gzipped fasta format. Please let me know if you need anything else!

Thank you,
Kylie

Kylie Shen, Bioinformatician
Eclipse BioInnovations
kylie@eclipsebio.com

On Aug 10, 2019, at 7:52 AM, 陈鹏达 <pandaman5555@qq.com> wrote:

Hi Kylie,

Thank you so much for your helpful answers.
I tried to use repbase for removing repetitive elements recently, but

unfortunately, I found the resource was required subscription. And our institution seemed not to be one of the subscribed institutions.
Did you have the mouse-specific version of repbase? Would you mind sending me if the documents wasn't very large?

Thank!
Pengda

------------------ 原始邮件 ------------------

发件人: "Kylie Shen"<kylie@eclipsebio.com>;
发送时间: 2019年8月9日(星期五) 凌晨5:28
收件人: "陈鹏达"<pandaman5555@qq.com>;
抄送: "Eric Van Nostrand"<eric@eclipsebio.com>;"肖昌春"<cxiao@xmu.edu.cn>;"alex"<alex@eclipsebio.com>;"廖堃玉"<779223993@qq.com>;
主题: Re: some bioinformatic problems from Changchun Xiao lab

Hi Pengda,

I've addressed your questions below:

1. Which step are you running FastQC on? In our data, we generally see that the GC content is high before the adapter trimming step, but looks better after adapter trimming and removing repetitive elements.
2. We generally recommend 50-75 nucleotides for single end sequencing. In general, the CLIP fragments are typically only ~30nt long, but we haven't seen a problem with 50 or 75nt reads (as adapter trimming removes readthrough sequences sufficiently).
3. Yes, you should run the pipeline using only R1 from your paired end data, and treat those reads as if it were a single end experiment. You can still run samtools view or samtools sort to analyze the alignments for single end sequences. Some of the samtools sorting is simply for later steps that require an indexed .bam file.
4. Yes, using samtools sort rather than sortSam from GATK is fine

Please let me know if you need any clarification on those answers, or if you have any other questions!

Thank you,
Kylie

Kylie Shen, Bioinformatician
Eclipse BioInnovations
kylie@eclipsebio.com

On Aug 7, 2019, at 10:52 AM, 陈鹏达 <pandaman5555@qq.com> wrote:

Hi Kylie,

I got some question during my recent analysis.
1. In our data, I found GC content was higher than the standard when I ran Fast.
2. As you mentioned, Single end is good enough for clip seq. Dose it mean the length of our read can not be too long,   otherwise SE can't get a good coverage.
3. If we used the reads with umi for analysis, can I just consider these reads as SE? Then I don' t need to do samtools view to sort single end sequence?
4. I was not familiar with GATK, so I try to use samtools sort command to replace the Samsort program from GATK. Is it ok?

Please take some time to give me some response if you got a time.
Thank you!
Pengda

---Original---

From: "Kylie Shen"<kylie@eclipsebio.com>
Date: Thu, Aug 1, 2019 01:05 AM
To: "陈鹏达"<pandaman5555@qq.com>;
Cc: "Eric Van Nostrand"<eric@eclipsebio.com>;"肖昌春" <cxiao@xmu.edu.cn>;"alex"<alex@eclipsebio.com>;"廖堃玉" <779223993@qq.com>;
Subject: Re: some bioinformatic problems from Changchun Xiao lab